

Day 10: Docker Volumes and Docker Networking with Commands

Docker Volumes

A **Docker Volume** is used to store data permanently outside a container. Even if a container is deleted, the data stored in the volume remains safe. Volumes are commonly used for databases, application files, and backups.

Common Docker Volume Commands

Create a Docker Volume

```
docker volume create myvolume
```

List All Volumes

```
docker volume ls
```

Inspect a Volume

```
docker volume inspect myvolume
```

Run a Container with a Volume

```
docker run -d -v myvolume:/app/data nginx
```

Remove a Volume

```
docker volume rm myvolume
```

Remove All Unused Volumes

```
docker volume prune
```

Advantages of Docker Volumes

- Persistent data storage
 - Data remains after container deletion
 - Easy backup and recovery
 - Volumes can be shared between containers
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Docker Networking

Docker Networking allows Docker containers to communicate with each other and with external networks. Docker supports several network types such as Bridge, Host, Overlay, None, and Macvlan.

Common Docker Network Commands

List Available Networks

```
docker network ls
```

Create a New Network

docker network create mynetwork

Inspect a Network

docker network inspect mynetwork

Run a Container on a Specific Network

docker run -d --network mynetwork --name web nginx

Connect a Running Container to a Network

docker network connect mynetwork web

Disconnect a Container from a Network

docker network disconnect mynetwork web

Remove a Network

docker network rm mynetwork

Advantages of Docker Networking

- Enables communication between containers
- Provides network isolation
- Supports multi-container applications
- Improves security and scalability

Conclusion

Docker Volumes provide persistent storage for container data, while Docker Networking enables secure communication between containers. Commands such as **docker volume create**, **docker run -v**, **docker network create**, and **docker network connect** are commonly used to manage storage and networking in Docker applications.



